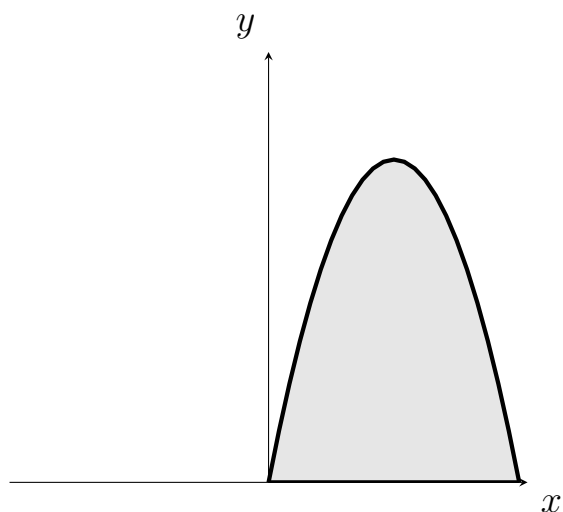


6.3 Volumes by Cylindrical Shells

Rotate the region enclosed by the x -axis and the parabola $y = 3x - x^2$ around the line $x = -1$.

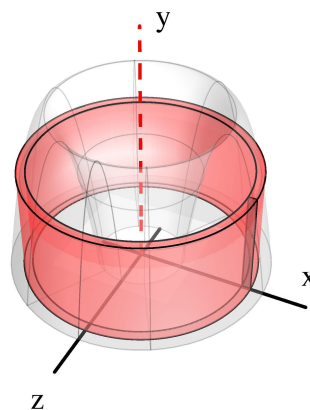
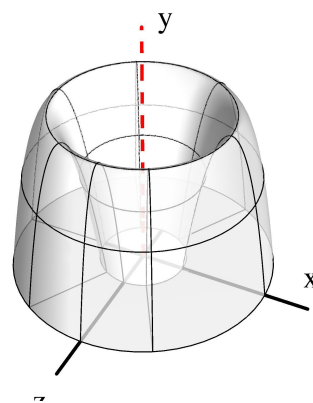
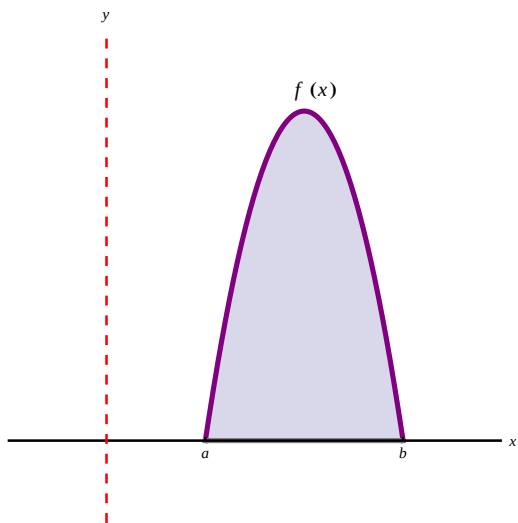


Suppose we wish to find the volume of this solid.
What method would we have to use?

Why might this be challenging?

In the disk and washer method, the cross-sections taken were _____ and _____ to the axis of rotation. We will introduce a new method, using a new shape for cross-sections, which are taken _____ to the axis of rotation.

Solids of Revolution: The Shell Method



Volume by Shells for Rotation about the x -axis

The formula of the solid generated by revolving the region between the x -axis and the continuous function $y = f(x) \geq 0$ where $L \leq a \leq x \leq b$ about a vertical line $x = L$, is

$$V =$$

The volume of the solid of revolution sliced into cylindrical shells of radius, $r(x)$, and height, $h(x)$, is

$$V =$$

Where does this formula come from? Let's derive it!

Example 1

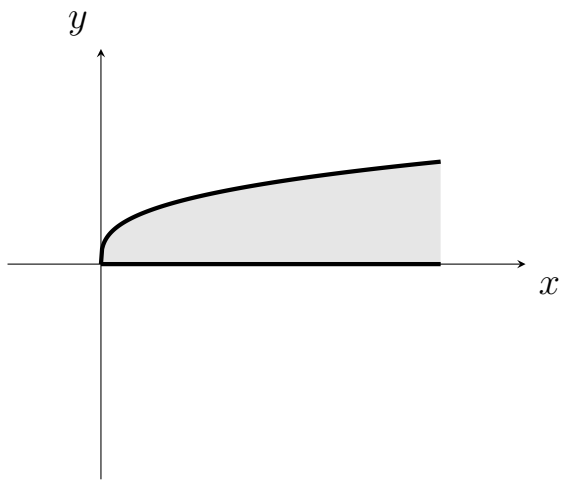
Set up an integral to find the volume of the solid obtained by rotating the region bounded by $y = 2\sqrt{x-1}$ and $y = x-1$ about the line $x = 6$.

Finding the Volume of a Solid of Revolution-Shell Method

1. Sketch the region that is being rotated.
2. Draw the _____.
3. Derive a formula for the _____ of the shell. The shell _____ can be found by drawing a line parallel to the axis of rotation.
4. Derive a formula for the _____ of the shell. The shell _____ is the distance from the axis of revolution.
5. Determine the variable of integration. Integration is performed with respect to the axis that is parallel to the slices.
6. Determine the _____ of integration. They must match the variable of integration!
7. Integrate. Remember units in your final answer when applicable.

Example 2

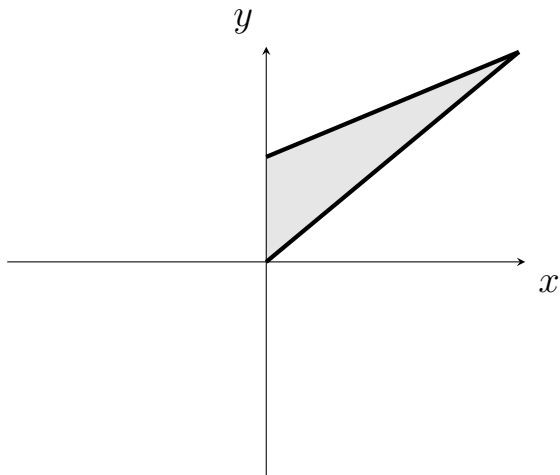
Set-up an integral to find the volume of the solid obtained by rotating the region bounded by $y = \sqrt[3]{x}$, $x = 0$, $x = 8$, and the x -axis about the line $y = -2$.



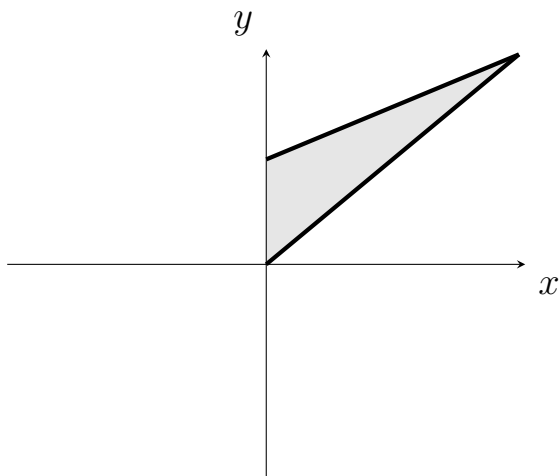
Example 3

Use an appropriate method to set-up an integral to find the volume of the solid generated by revolving the region bounded by $2y = x + 4$, $y = x$ and $x = 0$ about

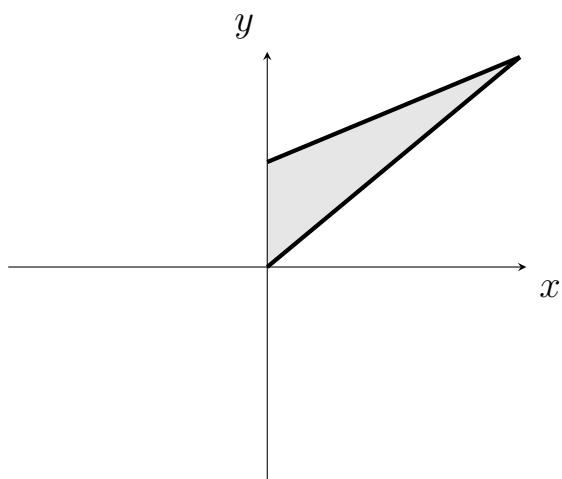
(a) the x -axis



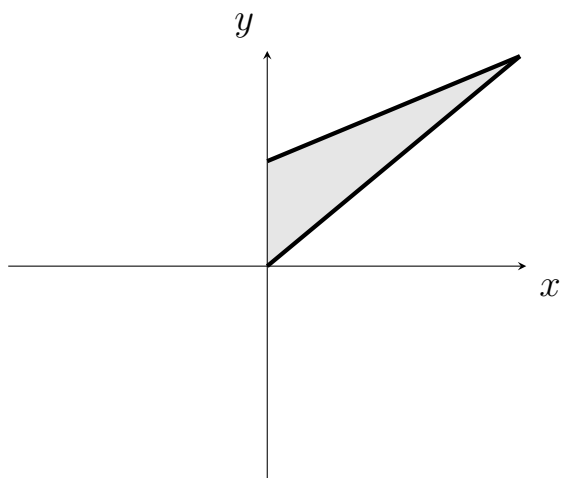
(b) the y -axis



(c) the line $x = 5$



(d) the line $y = 8$



Challenge Example

Students are asked to set up an integral that represents the volume generated by rotating the region bounded by the curves $y = 4x - x^2$ and the x -axis about $x = 7$.

A student proposes the following integral:

$$V = \int_0^4 \pi [(4x - x^2)^2 - (7)^2] \, dx$$

Is their answer correct?

If their answer is correct, then explain what method they are using and what the integrand represents.

If their answer is incorrect, then explain why. Your explanation should include the correct answer.